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Project Title:	VIBE Technology for Microgravity Heat Transfer Applications		
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Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
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Zip Code:	30332-0405	**Congressional District:**	5
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Task Description:	Thermal management is an important subsystem in many devices and technologies used in a microgravity environment. The increased power requirements of new Space technologies and missions mean that the capacity and efficiency of thermal management systems must be improved. Our work addresses this need through the investigation and development of a direct liquid immersion heat transfer cell for microgravity applications. The device is based on boiling heat transfer enhanced by two fluidic technologies developed at Georgia Tech. The first of these fluidic technologies is called vibration-induced bubble ejection. Here, a vibrating diaphragm is used to vibrate a vapor bubble on a heated surface. With driving frequencies of a few hundred Hertz, violent oscillations of the vapor/water interface will occur that can result in small bubbles being ejected from the larger bubble and, simultaneously, the collapse of the vapor/water interface against the solid surface. Both effects would be useful during a heat transfer process. Bubble ejection would force vapor bubbles back into the cooler liquid so that they can condense. Interfacial collapse would tend to keep the hot surface wet thereby increasing liquid evaporation and heat transfer to the bulk liquid. When the diaphragm is vibrated at higher frequencies (1~10 kHz), small-scale capillary waves appear on the surface of the bubble near the attachment point on the diaphragm. The vibration produces a net force on the bubble that pushes it away from the diaphragm. As a result, the bubble detaches from the solid surface and is propelled into the bulk liquid. The benefit of the force in a boiling process would be to push vapor bubbles off the heated surface, thus helping to keep the surface wet and to increase the heat transfer. The second fluidic technology to be employed in this work is a synthetic jet. The jet is produced using a small, sealed cavity with a sharp-edged orifice on one side and a vibrating diaphragm on the opposite side. The jet is formed when fluid is alternately sucked into and then expelled from the cavity by the motion of the diaphragm. This alternating motion means that there is no net mass addition to the system. Thus, there is no need for input piping or complex fluidic packaging. This jet can be used to push vapor bubbles off of the heated surface. The efficiency of boiling heat transfer is hindered in microgravity because there is no inherent mechanism to remove vapor bubbles from the heated surface (this is the role of buoyancy on Earth). To circumvent this problem, the fluidic technologies described above will be used to develop a nucleate boiling heat transfer cell for efficient, high-flux cooling in microgravity. The cell will be composed of a liquid-filled cavity with one surface attached to the surface to be cooled and with another surface attached to an array of heat transfer fins or some other device that serves to reject the heat to an outside environment. An array of vibration actuators will be designed and fabricated on or near the hot surface. These actuators will aid nucleate boiling in the cell by vibration-induced bubble ejection from larger vapor bubbles, vibration-induced collapse of vapor bubbles against the heated surface, and the direct removal of vapor bubbles from contact with the heated surface. Synthetic jets placed inside the cell will produce a flow that will transport vapor bubbles away from the heated surface into the cooler bulk liquid. Since the synthetic jets operate on time scales that are much shorter than the characteristic thermal time scale of the flow, they will also substantially enhance small-scale mixing within the liquid, thus enhancing the condensation of vapor bubbles back into the liquid and the heat transfer out of the cell. The primary objectives of this research are as follows: 1) Develop, design, and construct a nucleate boiling heat transfer cell based on the vibration-induced bubble ejection and synthetic-jet technologies. 2) Examine the formation, spreading, and detachment of vapor bubbles from a heated surface during boiling in microgravity. 3) Determine the influence of vibration of the heated surface on the spreading, merging, and detachment of vapor bubbles from the surface. 4) Determine the vibration conditions that maximize the interfacial motion of a vapor bubble and that ensure its collapse against the heated surface. 5) Determine the conditions for the effective use of forced convection induced by synthetic jets to remove vapor bubbles from the heated surface. If the fluidic technologies described above behave as expected, the simplicity and scalability of the resulting heat transfer cell would make it a very attractive method for thermal management in microgravity. In addition, the technology may also be transferable to Earth-based, boiling applications and provide improved performance over normal pool-boiling technologies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	A channel boiling heat transfer cell based on a synthetic jet technology has been demonstrated. A synthetic jet is produced by a submerged piezoelectric actuator vibrating at about 7 kHz. The actuator was placed on one side of a small rectangular channel and the jet was directed to impinge normally on the opposite heated surface. The jet removed the vapor bubbles that formed on the heated surface during nucleate boiling and forced them back into the cooler bulk liquid, where they condensed. A small cross flow within the cell and a constant temperature bath were used to maintain the background temperature of the cell at a constant value. The synthetic jet produced heat fluxes in the cell of 121 W/cm2 at a surface temperature of 111°C, an increase of 236% compared to a traditional channel boiling situation where the jet was turned off. The jet was even more effective (300%) when the inlet temperature of the water flowing through the cell was increased, although the total heat flux decreased. These increases in heat transfer were obtained at the expense of only 1 W of input power to the actuator. Using another heater design, the largest increase in heat flux with the synthetic jet occurred near 120°C. Beyond this point, the effectiveness of the jet declined until the surface temperature of the heater reached 146°C, at which point the heat transfer from the surface was approximately 300 W/cm2 and the jet no longer had any effect on the heat transfer. When the heat transfer cell was inclined at an angle of 45°, such that the water inlet was positioned below the water outlet, the influence of the jet was seen all the way to the maximum surface temperature of 146°C. The shape of the curve showed that the effect of the jet was diminishing and that it would probably become ineffective at a surface

	temperature somewhat above 146°C. The inclined cell was more effective for heat transfer than a horizontal cell because the inclination allowed the vapor bubbles and the warmer bulk fluid near the heater surface to move downstream through the action of buoyancy. The jet assisted the heat transfer by increased mixing of the incoming cooler fluid toward the heated surface. The net effect was that a heat flux of 333 W/cm2 at a surface temperature of 146°C was achieved when the heat transfer cell was inclined at 45°.
Bibliography Type:	Description: (Last Updated: 07/09/2021)
Dissertations and Theses	Tillery SW. "Enhanced Boiling Heat Transfer by Submerged, Vibration Induced Jets." M.S. Thesis, School of Mechanical Engineering, Georgia Institute of Technology, May 2004. , May-2004
Papers from Meeting Proceedings	Tillery SW, Heffington S, Smith MK, Glezer A. "Boiling Heat Transfer Enhancement by Submerged, Vibration Induced Jets." 9th Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems, Las Vegas, NV, June 2004. Proceedings: 9th Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems, Las Vegas, NV, 2004. p. 17–22. https://doi.org/10.1109/ITHERM.2004.1318247 , Jun-2004
Presentation	Tillery, S. W.; Heffington, S.; Smith, M. K.; and Glezer, A. "Boiling Heat Transfer Enhancement by Submerged, Vibration Induced Jets" ITherm 2004 Proceedings: 9th Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems, Las Vegas, NV, pp. 17–22, Jun-2004